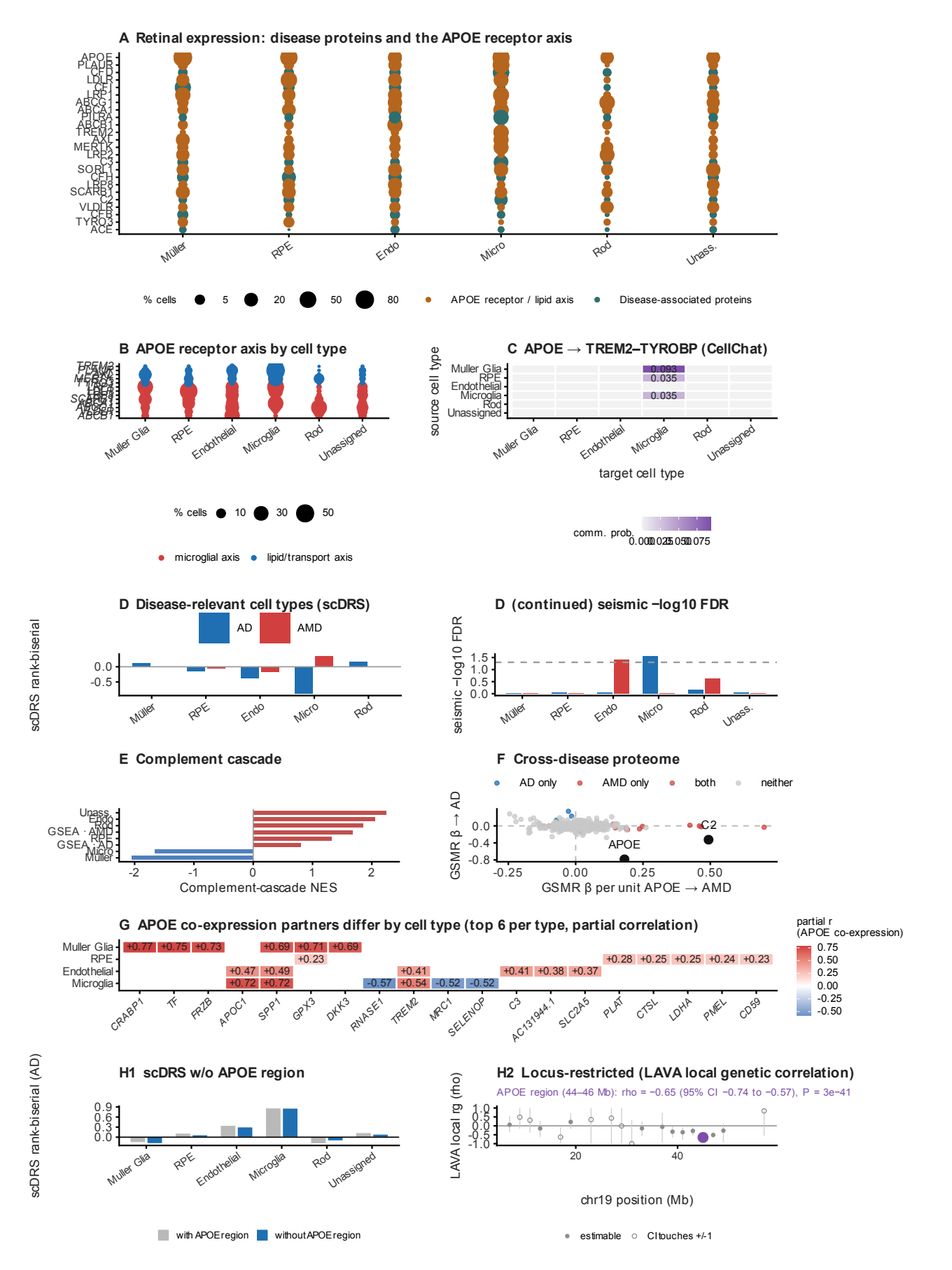

A Retinal expression: disease proteins and the APOE receptor axis
APOE
PLAUR
CFD
LDLR
CP
LRP1
ABCG1
ABCA1
PTPRA
ABCB1
TREM2
AXL
MERTK
LRP2
C3
SORL1
CFH
LRP8
SCARB1
C2
VLDLR
CR1
TYRO3
ACE
Muller
RPE
Endo
Micro
Rod
Unass.
% cells
5
20
50
80
APOE receptor / lipid axis
Disease-associated proteins
B APOE receptor axis by cell type
Muller Glia
RPE
Endothelial
Microglia
Rod
Unassigned
% cells
10
30
50
microglial axis
lipid/transport axis
C APOE → TREM2–TYROBP (CellChat)
source cell type
target cell type
Muller Glia
RPE
Endothelial
Microglia
Rod
Unassigned
0.093
0.035
0.035
comm. prob.
0.000
0.025
0.050
0.075
D Disease-relevant cell types (scDRS)
AD
AMD
scDRS rank-biserial
0.0
-0.5
Muller
RPE
Endo
Micro
Rod
D (continued) seismic −log10 FDR
seismic −log10 FDR
1.5
1.0
0.5
0.0
Muller
RPE
Endo
Micro
Rod
Unass.
E Complement cascade
Unass.
Endo
Rod
GSEA · AMD
RPE
GSEA · AD
Micro
Muller
-2
-1
0
1
2
Complement-cascade NES
F Cross-disease proteome
AD only
AMD only
both
neither
GSMR β → AD
0.0
-0.4
-0.8
C2
APOE
-0.25
0.00
0.25
0.50
GSMR β per unit APOE → AMD
G APOE co-expression partners differ by cell type (top 6 per type, partial correlation)
Muller Glia
RPE
Endothelial
Microglia
+0.77
+0.75
+0.73
+0.69
+0.71
+0.69
+0.23
+0.47
+0.49
+0.72
+0.72
-0.57
+0.41
+0.54
-0.52
-0.52
+0.41
+0.38
+0.37
+0.28
+0.25
+0.25
+0.24
+0.23
CRABP1
TF
FRZB
APOC1
SPP1
GPX3
DKK3
RNASE1
TREM2
MRC1
SELENOP
C3
AC131944.1
SLC2A5
PLAT
CTSL
LDHA
PMEL
CD59
partial r
(APOE co-expression)
0.75
0.50
0.25
0.00
-0.25
-0.50
H1 scDRS w/o APOE region
scDRS rank-biserial (AD)
0.9
0.6
0.3
0.0
Muller Glia
RPE
Endothelial
Microglia
Rod
Unassigned
with APOE region
without APOE region
H2 Locus-restricted (LAVA local genetic correlation)
APOE region (44–46 Mb): rho = −0.65 (95% CI −0.74 to −0.57), P = 3e−41
LAVA local rg (rho)
1.0
0.5
0.0
-0.5
-1.0
20
40
chr19 position (Mb)
estimable
CI touches +/-1